Eduardo Gracidas Reyes

Biomedical Engineer · M.Sc. Computer Science student

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PROFILE

Biomedical Engineer (UDLAP, *summa cum laude*) and fully funded M.Sc. Computer Science student at Memorial University of Newfoundland. My research combines instrumentation, biomechanics and computing, with a focus on insect-based biorobotics: I built a controlled-environment treadmill platform to study the locomotion of the Madagascar hissing cockroach, published as an **Editor's Choice** article in *Biology* (MDPI, Q1). I also develop software with JavaScript/Node.js, MongoDB and Python.

EDUCATION

Sep 2026 – present
M.Sc. Computer Science
Memorial University of Newfoundland, St. John's, Canada
Fully funded scholarship. Co-supervised by Dr. Oscar Meruvia Pastor and Dr. Jay Henderson. Current courses: Applied Algorithms, Software Engineering, Data Visualization.

2022 – 2026
B.Sc. (Honors) Biomedical Engineering — *Summa Cum Laude*
Universidad de las Américas Puebla (UDLAP), Puebla, Mexico
UDLAP Honors Program (2023–2026). Research thesis on the thermal and light-dependent locomotion of *Gromphadorhina portentosa* for biorobotic applications.

PUBLICATIONS

Journal article

> **Gracidas-Reyes, E.**; Díaz-Arango, G.; Vázquez-Leal, H.; Marroquín-Gutiérrez, F. (2026). Temperature-Dependent Tethered Locomotion Behavior in the Madagascar Hissing Cockroach Using a Controlled-Environment Treadmill Platform with Exploratory Illumination Assays. *Biology*, 15(12), 947. doi:10.3390/biology15120947. **Editor's Choice** **Q1 · IF 4.3**

Book chapters

- *Advances in the Design and Control of Cyborg Hissing Cockroaches: Machine Learning, Motion Tracking, and Temperature Control with a Perspective on Search and Rescue Operations.*
- *Adquisición de datos para el análisis de la locomoción de Gromphadorhina portentosa bajo distintas condiciones térmicas.*

RESEARCH EXPERIENCE

Sep 2025 – Mar 2026
Social service — INAOE, Electronics area
Instituto Nacional de Astrofísica, Óptica y Electrónica, Puebla, Mexico
- Characterization of *Gromphadorhina portentosa* under environmental and electrical stimuli for its use as a biorobotic system.
- Design and implementation of a controlled environment for temperature and light.

May – Aug 2025
Professional practices — INAOE, Electronics area
Under the direction of Dr. Luis Hernández Martínez
- Adapted specimens to controlled experimental conditions.
- Implemented an experimental environment with programmable temperature, humidity and light; results published in *Biology*.

Ongoing
Cyborg Insects Initiative (C2I)
Insect-based biorobotics
- Machine learning for stimulating cyborg cockroaches while avoiding habituation.
- Webcam + QR-code videogrammetry for motion tracking without heavy electronic backpacks.
- Characterization of stimulation parameters for efficient movement control.

HONOURS & AWARDS

2026 Editor's Choice article — *Biology* (MDPI)
2026 Fully funded M.Sc. scholarship — Memorial University of Newfoundland
2026 *Summa cum laude* — B.Sc. Biomedical Engineering, UDLAP

2025 3rd place — Silldromo 2025, biomedical applications from biosignals, INAOE
2025 1st place, Junior category — Expo CEM 2.0, UDLAP
2025 Winner — IICNNCD, cyborg-cockroach data acquisition and statistical modelling
— 2nd place, poster competition — II Congreso Nacional de Ciencia de Datos
2026 Recognition — Colegio Ángeles
2023 Member — UDLAP Honors Program

CONFERENCES & PRESENTATIONS

- International Conference on Ontologies and Knowledge Graphs (ICOKG 2025), BUAP — presenter
- Congreso Nacional de Ingeniería Biomédica (CNIB 2025), SOMIB — poster "Sistema de análisis biomecánico"
- 7ª Jornada de Mujeres en la Ciencia, BUAP (2025) — poster "Sistema de monitoreo de señales biomecánicas"
- II Congreso del Programa de Honores, UDLAP — panelist
- Expo CEM 3.0 (Fall 2025), Expo CEM 2.0 (Spring 2025) and Expo CEM (Fall 2024), UDLAP

SELECTED PROJECTS

- **Semi-portable Smart Pillbox for polypharmacy patients.** 21-compartment automatic dispenser, audiovisual alarms, TFT display and a cross-platform mobile app (SmartDose).
- **IoMT system for surgery and remote monitoring.** Control interface with live video and remote ECG for surgical support and remote follow-up.
- **Feature extraction for Parkinson disease diagnosis.** Physiological-signal processing to extract features supporting clinical diagnosis.
- **3D reconstruction of CT images.** Volumetric reconstruction of CT slices with axial, coronal and sagittal views.
- **Stock-trading simulation (Software Engineering course).** Team-based software engineering project at Memorial University.

CERTIFICATIONS & COURSES

- Focus on Peer Review — Nature Masterclasses (2025)
- AI in Global Health Research, MSC1125H — University of Toronto (2025)
- Startups in the Medical Sciences, MSC1122H — University of Toronto (2025)
- Wearable Sensors for Cardiorespiratory Monitoring — IEEE (2025)
- Bachelor's Master Classes — Boston Scientific (2025)
- Diploma in Neurolearning, 100 h — Universidad Anáhuac Puebla (2025)
- AI in Medicine, MSC1114H — University of Toronto (2023)
- Microsoft Office Specialist: Excel 2016 (2018)

SKILLS

Software	JavaScript, Node.js, Express, ViteExpress, REST APIs, OpenAPI, JSDoc, automated testing, HTML/CSS, software architecture
Data	MongoDB Atlas (native driver), Plotly.js, data visualization, market-data APIs
Scientific	Python, Jupyter, signal processing, mixed-effects models, non-parametric statistics, computer vision, machine learning
Biomedical	Instrumentation, embedded systems, microcontrollers, biosignal acquisition, thermal imaging, environmental control
Tools	Git, GitHub, scientific writing, peer review

LANGUAGES

Spanish (native) · English (professional working)

RESEARCH INTERESTS

Biorobotics and biohybrid systems · insect neuroethology and locomotion · biomedical instrumentation and biosignals · computer vision and motion tracking · machine learning for control · data visualization